

The Fibonacci Neighborhood

beats Adjacent and Dynasearch from $n = 200$

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At $n = 200$, $m = 20$, $t/l = 5$ s, Tabu Search:

13.19%

Adjacent

single swap, $O(mn)$

12.71%

Fibonacci

this work

13.20%

Dynasearch

segment swaps, $O(mn^3)$

0.48 pp better than Adjacent, 0.49 pp better than Dynasearch at quadratic per-iteration cost

The Fibonacci move

A composite move is a subset $S \subseteq \{1, \dots, n-1\}$ of adjacent-swap positions with **no two consecutive selected**:

$$i \in S \implies i + 1 \notin S.$$

All selected swaps $\pi_i \leftrightarrow \pi_{i+1}$ are applied **simultaneously**.

swapswap swap

0 1 2 3 4 5 6 7

Why "Fibonacci"? The number of valid subsets in a permutation of length n equals the Fibonacci number F_{n+1} .

Algorithm – one DP pass

1. Recompute $C_{\max}(\pi^{(i)})$ for all i

$O(mn)$ each

2. DP picks optimal non-overlapping subset

$$\text{dp}[i] = \min(\text{dp}[i + 1], \Delta_i + \text{dp}[i + 2])$$

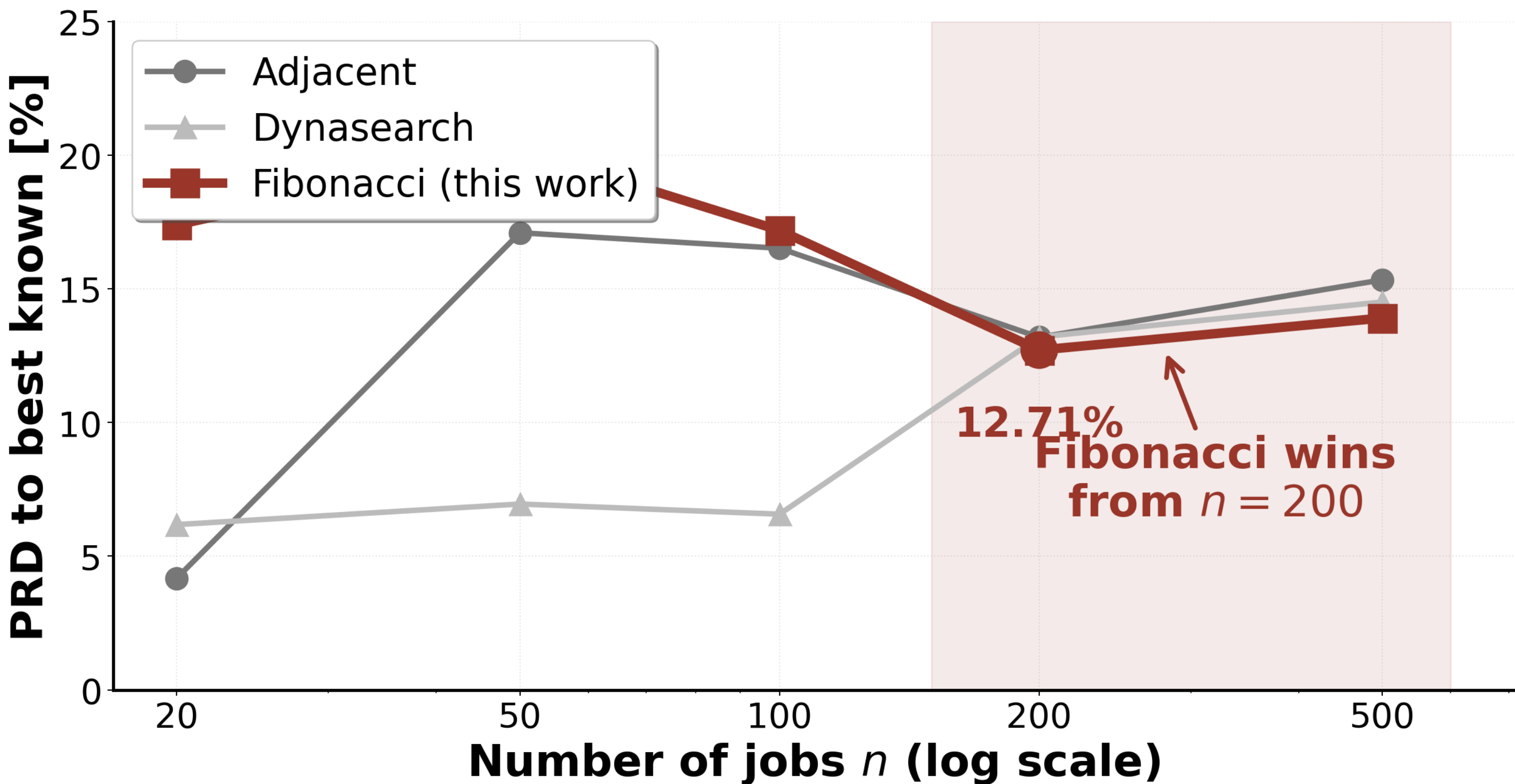
3. Apply all selected swaps at once

Neighborhood	Candidates	Per-iter cost
Adjacent	$n - 1$	$O(mn)$
Fibonacci	$n - 1 + \text{DP}$	$O(mn^2)$
Dynasearch (rec.)	$O(n^2)$ segments	$O(mn^3)$

F_{n+1} valid moves explored at **quadratic** cost – the sweet spot between cheap (Adjacent) and deep (Dynasearch).

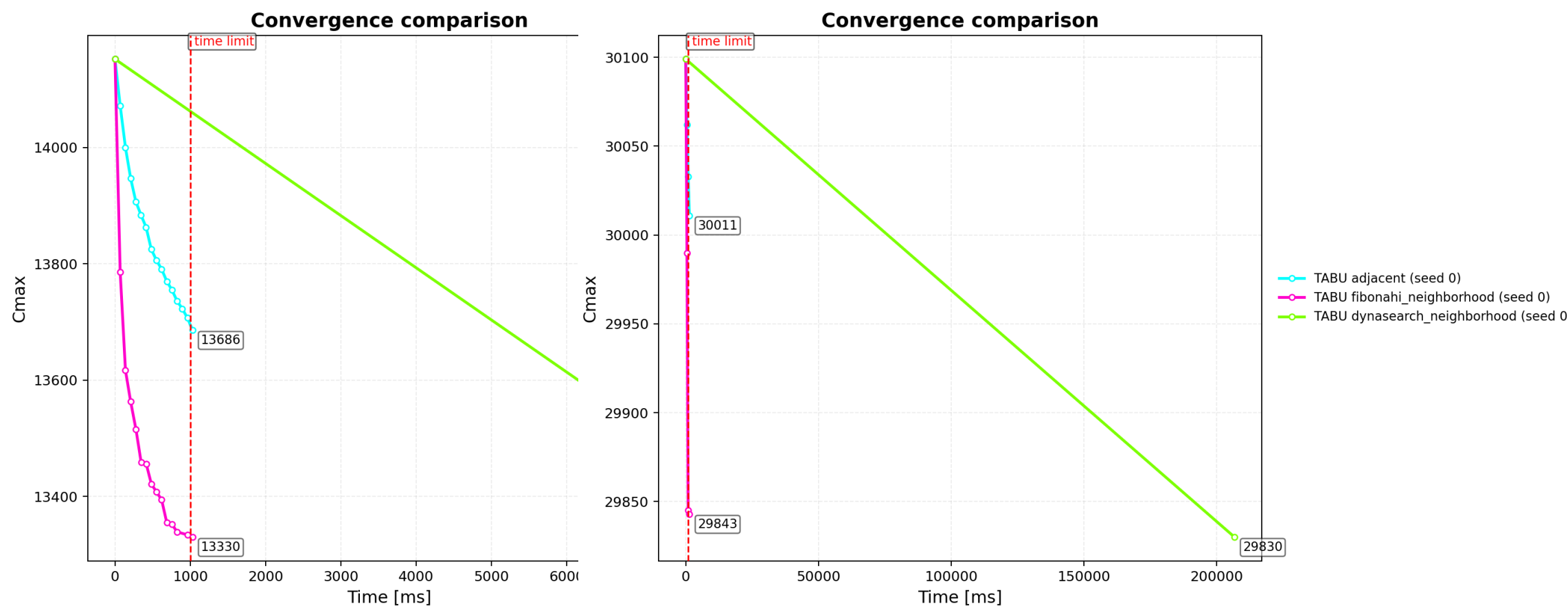
Where Fibonacci wins

Mean PRD across Taillard instances — Tabu Search, $t/l = 5$ s



Crossover at $n = 200$. Below $n = 100$ Dynasearch's deep moves dominate. From $n = 200$ Dynasearch's $O(mn^3)$ cost makes it unable to finish a single iteration within the budget – Fibonacci's cheaper compound move pulls ahead and stays competitive at $n = 500$.

Convergence – $t/l = 1$ s



tai200_20

tai500_20

Fibonacci (**magenta**) drops $C_{\max}$ in the first ~ 100 ms. Dynasearch (orange) barely finishes one iteration within the budget.

Take-aways

Default for $n \geq 200$: Fibonacci, period.

SA reheating rescues the composite move from plateaus once all $\Delta_i \geq 0$.

Dynasearch at $n = 500$ can need $200\times$ the budget for a single iteration – skip it under tight t/l .